

Genomic analyses reveal the origin of domestic ducks and identify different genetic underpinnings of wild ducks.

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Abstract

Background: Domestic ducks are considered to have been tamed from the mallard or a descendant of the mallard and the spot-billed duck. Domestic ducks show remarkable phenotypic variation in morphology, physiology and behaviour. However, the molecular genetics of the origin and phenotypic variation of ducks are still poorly studied.

Results: We presented mallard and spot-billed genomes and performed whole-genome sequencing on eight domestic duck breeds and eight wild duck species and aligned to the most widely used assembly, BGI_duck_1.0 Pekin. Surprisingly, analyses of these data support a model in which domestic ducks may have diverged from their closest wild lineage (mallard ducks and spot-billed ducks) before the splitting of the two wild species. The subsequent formation of mallard and spot-billed ducks from wild lineage was further indicated by the genomic landscape, which exhibited characteristics of recent speciation. Meanwhile, the domestic lineage population decreased through the last glacial period (LGP). A scan of wild duck genomes compared with domestic duck genomes identified numerous loci that may have been affected by positive selection in ancestral wild ducks after their divergence from domestic lineages. Function analyses suggested that genes usually affecting organ development and energy metabolism may involve long-distance flight ability. Further selective sweep analyses identified two genes associated with egg production and three genes related to feeding behaviour under selection in domestic ducks.

Conclusions: These analyses unravel a distinct evolutionary pattern of duck domestication and the opposite evolutionary paths observed in two wild counterparts, thus providing a valuable resource for speciation studies and a novel perspective or important supplementary on animal domestication.

Key Words: Adaptation; Molecular Evolution; Domestication; Speciation;

48 Population genetics; Flightness of birds

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Introduction

Domestication and speciation (or population divergence) have drawn a lot of attention in evolutionary studies especially since the emergence of new genomic methods with the deluge of empirical sequencing data[1-5]. The speculative origins of domesticated animals are often based on prior assumptions, such as suggesting that this process began during the period of human civilization. And it implicitly implied that the domestic lineage occurred in this period, which was split from their wild relatives directly. Genome sequencing studies in domestication research usually identify genes that exhibit clear genetic differences in domestics, but poorly resolved the demography. According to such assumptions, which propose that domestications are initiated by human and supposed to occur with bottlenecks, demographic simulations with longer divergence time between domestics and wild relatives are often discarded [1, 6]. Some research found that assumptions based on such understanding are incompatible with genetic data, historical and archaeological records[1, 7-9]. The assumptions have even been referred to as a mis-conception, and the in-concordant in estimated time were also mentioned to be influenced by mutation rates and other parameters' bias [1, 8, 9]. This may also be attributed to the inherent challenges faced by current simulation methods in accurately inferring the very recent history, particularly in the scale of human civilizations (e.g. PSMC's limited to infer<10Kya) [10-13].

There are currently more than 140 wild duck species worldwide, most species can hybridize with each other and two closest species (*Anas platyrhynchos* we referred as mallard in this paper and *Anas zonorhyncha*, refer as spot-billed) are thought to be the progenitors of the domestic duck, that is mainly based on their similar outlooks, habits, and distributions [14-16]. The phylogeographic of a recent radiation in the Aves:*Anas* were inferred from genomic transect and mallard duck diverged from its

closest relative (spot-billed duck) recently was further determined using the mtDNA control region and ODC-6 of nuclear DNA [17].

Despite the whole genome variations of duck have been characterized to explore the genetic variations associated with specific domestic traits. Whereas, they were still conducted under the prior assumption of some questionable hypothesis [8], and there are broad phenotypic differences between domestic ducks and their wild relatives deserving attention, such as loss of flight and migration ability for most domestic breeds, reduced brain size, and increased body size and fertility. Although researches on flightless, including steamer ducks, have revealed their diverse genetic mechanisms [12, 13], historical context have been proposed to be important in understanding functional changes and its ecology factors [18, 19]. However, the demographic inference suffered from lacking of solid history information, large amount unvalidated details, complex history and no robust mutation rates. The historical relationships of ducks, like many other domestic animals, are still unclear [20-24].

In this research, we aligned wild and domestic individuals of resequencing data to the most widely used assembly, BGI_duck_1.0 pekin duck. Through determining the phylogenetic topology of mallard, spot-billed and domestic lineages, we found our case can bypass the influence of the mutation rate and other simulation bias on inferring a specific divergence time since it could be derived by their splitting orders, thus making ducks an excellent case to learn the domestication process. In the end, we characterized the genomic divergence of mallard and spot-billed and the introgression effects to illustrate the very recent divergence time between the two wild ducks.

Results

We report genome drafts of the mallard (chromosome-level) and spot-billed and

compare them to the genome of pekin ducks and other birds (Additional file 1 section 1-3; Fig. S1-S10. Table S1-S10). We also generated a total of 1,219 Gb whole-genome resequencing data of 35 wild ducks from eight species, 114 domesticated ducks from six Chinese breeds and two commercial breeds (Cherry Valley and Campbell ducks), and 10 birds from Muscovy ducks and Taihu geese each as outgroups (Additional file 1 section 4; Fig. S11; Table S11). Variations were identified from different population combinations according to their sequencing coverages and specific requirements of different analyses, resulted in 44 million SNPs that were identified in the 10 duck populations (eight domestic duck breeds and two wild duck populations), and 20,676,377 and 42,012,846 SNPs were detected in domestic duck and two wild duck populations (the mallard and spot-billed), respectively, with 18,093,115 shared SNPs. Additionally, mallard and spot-billed ducks shared a comparative number of SNPs with domestic ducks (Additional file 1 section 5.2; Fig. S12a, b).

Domestic ducks represent an old lineage may predates the divergence of mallards and spot-bills

To explored the genetic relationships of domestic and wild ducks, Principal Component Analysis (PCA) were firstly performed using genome-wide SNPs from 27 Shaoxing and 27 Pekin ducks (individually sequenced domestic ducks) and 35 wild ducks. Expectedly, the domestic ducks (Shaoxing and Pekin ducks) clustered with mallard and spot-billed while clearly separating from other wild species (Fig. 1a; Additional file 1 section 5.4). Such genetic relationships were also identified from additional PCA, and IBS (identical by state) distance tree analyses of those 89 individuals (Fig. 1b; Fig. S13-S14). These results suggest that the mallard and spot-billed have a closer genetic relationship with domestic populations than other wild

species, which is consistent with our general recognition.

We then asked whether domestic ducks originated from the mallard or the descendants of the mallard and spot-billed. We performed a 3-population test for admixture by taking the mallard and spot-billed as ancestral populations and Pekin and Shaoxing as target populations through a Block Jackknife analysis that corrected for linkage disequilibrium among SNPs [25, 26]. Whereas, this effort obtained significantly positive scores for these four groups ($f_3=0.007279$, Z score=23.834; Table 1; Additional file 1 section 5.4.3) that did not suggest a mixed origin of domestic duck from mallard and spot-billed ducks. However, the mallard specific SNPs (polymorphism present only in mallard and ancestral allele fixed in spot-billed) and spot-billed specific SNPs were almost equally distributed in each of the 8 domestic breeds, which implied equally close kinship with domestic ducks, thus also conflict with single origin assumption (Fig. S12b).

Although we didn't find mixture origin evidence from f_3 -statistics, the genetic structure of four individual sequenced population (Pekin, Shaoxing, mallard and spot-billed) with K (the number of ancestry components) ranging from two to seven, provided new clues to this conflict [27](Additional file 1 section 4, 5.4). First, structure analysis indicated that domestic ducks had a relative homogeneous genetic background, whereas wild ducks showed a heterogeneous genetic background; this outcome is consistent with the PCA, nucleotide/population diversity and LD (domestic breeds present a low LD background than wild, whereas the elevated LD of adjacent distance in domestics may be caused by intense artificial selection during breeding and is consistent with the strong selection in Pekin than Shaoxing breeds) (Fig. 1; Fig. S12-S14). More importantly, the wild ancestor duck started to divide into mallard and spot-billed when K was increased to four, and it followed the separation

between the wild ducks and domestic ducks when $K=2$ which had the lowest cross-validation error (Fig. 1d; Fig. S15). These analyses implied a deeper divergence time between domestic lineage and wild than that between mallard and spot-billed, instead of traditional hypothesis (Fig. 1e). F_{ST} -based (population differentiation) phylogenetic tree was also compatible with this inference, and together with PCA results indicated that Shaoxing and Pekin represent our domestic samples comprehensively (i.e. from distinct clusters; Fig. S13). This evolutionary model could also explain the positive f_3 and the comparative numbers of mallard/spot-billed specific SNPs observed in each domestic duck breeds. Probably because those SNPs that shared by the wild ancestors and domestic ancestors drafted into mallard or spot-billed specific SNPs equiprobably (Fig. 1c; Fig. S12b). Although the resolution of IBS tree and genetic structure seems couldn't distinguish mallard and spot-billed well, possibly due to their very recent divergence stage, overall analyses indicated that they were monophyletic (Additional file 1 section 5.4.2; Fig. S13d, S15b).

To further confirm the evolutionary relationships of mallard, spot-billed and domestic ducks, we then randomly selected 20,333 unlinked SNPs from all 10 duck populations that ancestral state was polarized using two outgroups for a demographic simulation using the diffusion approach (*∂a∂i*) (Methods) [28]. We estimated the divergence time between the domestic ducks (quantized all eight domestic breeds into one population) and two wild duck populations (mallard and spot-billed) using two simple but informative models, with one taking migration into account while the other did not (Fig. 2a; Additional file 1 section 5.5). One hundred bootstrap simulations of the two models showed that both divergence times between the two wild duck populations and domestic ducks were almost equal (100-300 Kya) and longer than the corresponding one between mallard and spot-billed ducks (approximately 70 Kya,

Fig. 2b-c; Fig. S16-S18; Table S12), supporting the inference that the ancestor of the domestic duck split from the ancestor of mallard and spot-billed ducks before the division of two wild ducks (Fig. 1e). Although $\partial a \partial i$ couldn't distinguish short-divergence from high-migration which brought a noteworthy consequence when introducing migration factors, the estimate T_s of model 2 increased and with two peaks corresponding to longer T_s accompanied by high migration rates/shorter T_s coupled with low migration rates, simulation results of both scenarios supported our previous inference (Fig. 1,2).

In addition, we performed diploid pairwise sequential Markovian coalescent (PSMC) analysis to reconstruct the demographic history and estimated the effective population size (N_e) change along time among the mallard, spot-billed, Pekin and Shaoxing duck individuals. This analysis suggested that ancestors of these four populations presented a comparative effective population size before 110 Kya (beginning of the LGP). However, ancestors of the two wild ducks and domestic ducks seemed to be under different demographic histories after the LGP. During this period, the effective populations of Pekin and Shaoxing duck ancestors decreased dramatically and continually. In contrast, the mallard and spot-billed duck ancestors maintain a relative high population size until LGM (Fig. 2d; Fig. S19). This observation is consistent with previous research [12] and the population structure. And ancient introgression may make the curve of wild ducks heterogenous. PSMC can also be informally used to infer divergent times when N_e trajectories that are overlapping diverge and move forward in time towards the present [9, 29] and we found the divergent time roughly coincided with the one inferred from $\partial a \partial i$. Given this indication, we adjusted demography model that $\partial a \partial i$ fitted above to allow population size change exponentially after divergence to test whether the tendency of population size change

in $d_{a\partial i}$ is consistent with N_e value estimated by PSMC. we found that the population size of domestic duck mostly decreased, whereas both wild duck species increased population size, in all simulations, after the split between the wild and domestic population. This result matched the one from PSMC (Additional file 1 section 5.5).

Very recent speciation characteristic of genome between mallard and spot-billed reflect higher phylogenetic affinity between the two wild species than with domestic

If our inference of the phylogenetic relationship captured a more real demographic history than traditional hypotheses (Fig. 2a-b). we could expect that the two wild species emerged very recently and the genomic features of speciation species (e.g. ‘genomic islands’) that is recently studied extensively should be observed. One report using mtDNA sequences supported very recent divergence of mallard and spot-billed [17]. We then tried to identify ‘genomic islands’ between mallard and spot-billed through the fixed difference (d_f) or whether ‘differentiation islands’ exist through F_{ST} (fixation index) for 40 kb bins respectively, which has been regarded as an indication of genome in speciation studies, and showed highly elevated divergence up to 50 times the genomic median d_f in two flycatcher species whose divergence time estimated to be 340 Kya~2 Mya [30, 31]. Using the same criterion, fewer d_f (4,386) and a larger proportion (80%) of all d_f contained in the “genomic islands” are distributed in smaller genomic regions (0.6% of duck genome) between mallard and spot-billed ducks than flycatchers (59,936 d_f in 2.7% flycatcher genome) (Additional file 1 section 5.6.2).

This sharp difference might because the two wild ducks are in an earlier stage of speciation than flycatchers’ case. Moreover, almost all “genomic islands” and d_f sites were located in scaffolds homologous to the chicken “Z” chromosome (Fig. S20). To

identify genomic divergence in autosome, we further estimated F_{ST} through genome and also found that scaffolds aligned to Z chromosome of chicken showed 4.5-fold higher F_{ST} than those homologous to autosomes average. This may be due to the fast-Z effect[32, 33] and can be regarded as consistent evidence supporting the reported inference that low recombination rate and N_e may lead to pronounced effects of linked selection [34]. In genome wide, F_{ST} between mallard and spot-billed distributed in a narrower range (from -0.0157 to 0.7785 with an average of 0.0493) than that (ranged from 0.088 to 0.8912 with an average of 0.21) between the wild ducks and domestic ducks (Fig. 4f). Whereas, there is still a small subset of regions showing significantly higher differentiation located on autosomes and we referred as ‘high differentiation’ islands, which regions on autosomes elevated F_{ST} four standard deviations higher than genomic average (Fig. 4; Fig. S20-S21 and Discussion section).

These high differentiation regions showed lower level of nucleotide diversity and SNP density than genomic backgrounds in both mallard and spot-billed, which is consistent with the features of heterogenous genomic landscape reported in many speciation cases, which is considered caused by linked selection [3, 34-36] (Additional file 1 section 5.6.2). It’s noteworthy that d_{xy} didn’t elevate and even reduced in most of the ‘differentiation islands’, which makes the higher F_{ST} appear to be caused by reduced diversity in ancestral population rather than independent selection in the two lineages [30, 35, 37, 38]. Furthermore, Tajima’s D in high differentiation islands was not reduced in either mallard or spot-billed, which is different from observations in other speciation cases [3, 37]. It could also serve as an evidence of more recent/ongoing speciation stage, where there haven’t been enough rare mutations deposited yet after strong linked selection due to too short divergence

time; this finding may also be an evidence of their very recent speciation (Fig. 4b; Fig. S21; Discussion section).

Ancestors of mallard and spot-billed might benefit from adaptive selections

Intriguingly, the trend of population size change differed between the wild lineage and domestic lineage after their divergence during the LGP. One possibility might be that natural selection granted the mallard and spot-billed ancestors, but not domestic duck ancestors, new trait advantages to adjust to environmental change after their split. Consequently, environmental changes led to a decreased population size of domestic ancestor but increased one of wild ancestor (Fig. 2d; Additional file 1 section 5.4.1). We detected positive selection in the early ancestors of these two wild populations after their divergence with the domestic lineage and identified a total of 434 putative selective sweep regions, with a total size of 10.86 Mb, using thresholds of 4 standard deviations from the mean (Fig. 3a-b; Fig. S22, S23; Methods) [39, 40]. Only 150k of these putative sweep regions were overlapped with regions subsequently detected by F_{ST} between wild and domestic, further indicating that this signal capture different property. Among them, 168 selective sweeps harboured 202 protein coding genes while the remaining sweeps did not contain annotated protein coding genes (Additional file 1; Table S13). Recent studies have indicated that positive selection in intergenic and intragenic regions was associated with phenotypic variation [5, 7, 40]. We then extended 180 kb in both the upstream and downstream regions that did not contain annotated genes and identified 162 closest genes in 156 of these extended regions.

Subsequently, we performed a Gene Ontology (GO) analysis of 306 genes (12 of them detected in domestic selection were removed, Methods) in or closest to (extended 180 kb in both upstream and downstream) the selective sweep regions

(Additional file 1) and highlighted 343 enriched categories related to 236 biological processes (Table S14, S15). Interestingly, the most frequent categories were related to a series of organs and systems, such as morphogenesis and development of the lung, heart, metanephros, aortic and cardiac muscle and functions of the blood circulatory system, respiratory system, peripheral nervous system and ventricular system development, and rhythmic excitation (Table 2). Because long-distance flight requires adaptive changes in body shape, energy metabolism and spatial navigation, we proposed that the selection of genes associated with the above functions might have enhanced the ability to fly long distances in the ancestor of mallard and spot-billed (Additional file 1).

We further investigated functions of genes in putative selective sweep regions by surveying published literature and listed several genes that might be related to flight activity. The selection of the two genes *DNAJA1*, which stimulates ATP hydrolysis ([41]Baaklini et al., 2012), and *MRPS27*, which aids in protein synthesis within mitochondria [42], might help supply increased energy for sustained flight. We also noted that *PARK2* was under selection when using a lower threshold ($S=-3$). *PARK2* promotes the autophagic degradation of dysfunctional depolarized mitochondria, potentially suggesting an adaptation to facilitate the degradation of damaged mitochondria caused by products of an elevated metabolic rate. Previous studies found that some genes play important roles in energy metabolism, such as ferrous iron transport (i.e., *SLC39A14*) or positive fever generation regulation (i.e., *PTGS2*) [43, 44]. Positive selection of these genes in the early mallard and spot-billed duck ancestors might have enhanced their ability to transport oxygen in blood and adapt to cold and hypoxic environments at high altitudes during flight. Interestingly, we found that some genes involving retina and forebrain regionalization may implicate

functions related to magnetic navigation ability (Additional file 1 section 5.7.1). In addition, we examined the functional activity of four mutations located in introns, UTRs (Untranslated Region) or downstream of *EFNA5*, *ISL1* and *MRPS27*. Three of the four genomic sites with wild-type sequences showed significantly higher gene expression than the domestic genotype in a luciferase activity assay (Fig. 3d; Fig. S24; Table S16).

Identification of selective sweep regions in domestic duck

To detect genomic regions under domestic selection, we searched the duck genome for regions with reduced heterozygosity (H_p) and increased fixation index (F_{ST}) using methods similar to the research of chicken and dog genome (Additional file 1; Fig. S25) [2, 5]. We Z-transformed the H_p and F_{ST} distributions of autosomes and sex chromosomes separately and focused on putatively selective sweep regions falling at least four standard deviations away from the mean ($Z(H_p) < -4$ and $Z(F_{ST}) > 4$) (Additional file 1 section 5.7.2; Fig. 3c; Fig. S26, S27). However, we found that only a small proportion of the selective sweep regions were predicted using both signals (Fig. S28; Table S18, S19), which may attribute to the long independent evolutionary histories since their split (100-300 Kya). Therefore, we focused on these 137 regions with extremely low levels of H_p (average length=63.6230 kb, average H_p =0.2100 for autosomes, 0.2470 for sex chromosomes) (Additional file 1). We found that 131 genes and 289 miRNAs were contained in 80 of these CDRs while 47 genes were closest to one of the 33 CDRs when we extended 180 kb both upstream and downstream of these 57 CDRs containing no annotated protein coding genes (Table S18). Further GO analyses indicated that 162 genes in or closest to these 137 CDRs were enriched for 145 biological process categories (p -value<0.05), which represented developmental signalling; the nervous system; lipid, carbohydrate and

protein metabolic processes; cell division; and immune responses (Table S20, S21). Among them, two genes (*NLGN1* and *PER2*) are involved in the circadian sleep/wake cycle or associated with egg production [45], three genes (*MRAP2*, *MC4R* and *RMI1*) are associated with the modulation of feeding behaviour or reducing food intake in response to dietary excess, and four genes (*HSD17B4*, *HEXB*, *B4GALT1* and *CYP19A1*) are related to sexual characteristics, male courtship behaviour or androgen metabolic processes ([46, 47] (Fig. S29).

Among these candidate domestication genes, we examined the functional activity of two mutations located in the introns of *GRIK1* and *LYST*. The domestic genotype of the *LYST* mutation showed significantly higher gene expression activity than that of wild genotype in a luciferase activity assay (Fig. 3d; Fig. S24; Table S16)

Absence of introgression evidence of mallard and spot-billed further supported their shallow divergence without being confounded by cross hybridization

Although introgression has been considered in our previous analyses and seems unlikely to affect our inference, the Anseriformes is known for its propensity to interspecific hybridize. To further investigate the evidence of introgression between mallard and spot-billed we stratified the genomic divergence values of each window by Δ DAF (derived allele frequency difference) or absolute Δ DAF between two populations bins. According to recent research on introgression issues in chimpanzee, the chance of derived alleles being introduced through gene flow increases with the frequency in the donor population. Gene flow should be introduced into receptor populations at low frequencies ([48]. Therefore, we may expect lower divergence values in these higher Δ DAF (absolute Δ DAF) bins because they represent a higher probability of introgression. We did find decreased genetic distance (F_{ST} , d_{xy}) between mallard/spot-billed and domestic ducks in higher Δ DAF bins (Fig. S30a). on

the contrary, divergence (F_{ST} and d_{xy} values between mallard and spot-billed ducks increased with the ΔDAF (Fig. S30a blue lines). This observation provided evidence of introgression between wild and domestic but not found between mallard and spot-billed. Furthermore, we found that the mean ΔDAF value did not show significant difference between high differentiation and background genomic regions; this finding is consistent with recent research that gene flow is not the major factor in the formation of genomic islands (Table S22) [37].

To further decipher the introgression effect and its influence on the evolutionary relationship inference of the domestic and the two wild ducks, we performed an ABBA-BABA test (assuming that the mallard, spot-billed, and domestic duck are P1, P2, and P3, respectively) using the Muscovy duck as an outgroup. The analysis suggested that the genetic relationship between mallard and spot-billed ducks is closer than that between the two wild species and the domestic duck (Fig. S31a). And we also found evidence of gene flow between the domestic and wild ducks that is all consistent with the above analysis (D-statistics: -0.2972; jackknife std: 0.4189; Fig. 1d, Fig. S15). To investigate whether gene flow has its effect on this phylogenetic tree, we used ΔDAF to test for evidence that the sharing of this derived allele is caused by introgression. We stratified the number of BBAA (ABBA, BABA) sites by the ΔDAF bins between mallard and spot-billed duck populations (spot-billed and domestic, mallard and domestic, respectively). Most BBAA sites were distributed in the lower ΔDAF bins (-0.4 ~ 0.4), which is consistent with our previous inference that majority of genome region showed lower levels divergence between mallard and spot-billed ducks. To eliminate the inhomogeneity distribution across the ΔDAF bins which is caused by phylogeny SNPs, we further measured the distribution of the BBAA ratio among all SNPs (i.e., polymorphic sites in spot-billed and domestic,

mallard and domestic, mallard and spot-billed for ABBA, BABA, and BBAA, respectively). If introgression exists between mallard and spot-billed ducks (mallard and domestic, spot-billed and domestic), then the proportion of BBAA sites would likely increase with the Δ DAF (or absolute Δ DAF if negative). We did find that the proportion of BABA and ABBA sites increased with the Δ DAF, indicating that the derived allele sharing between wild (mallard or spot-billed) and domestic populations is partly due to introgression, which is consistent with previous analysis (Fig. 1d; Fig. S30). Whereas the distribution of the BBAA proportion across Δ DAF bins showed a different pattern than the BBAA rate, which showed only a small difference across Δ DAF bins, indicating that there is lack of introgression between mallard and spot-billed ducks, and implying that the major reason for excessive derived allele sharing is due to phylogeny rather than introgression (Fig. S31b).

Discussion

Animal domestication are very valuable research area and the research has been vastly boosted by the development of sequencing technology in decades[1, 2, 7, 9, 49]. Whereas, it has been hampered recently due to the incompatibility between traditional hypotheses and the genetic data or other evidences such as rare domestication episodes, hard to detect bottleneck, diffuse process and complex interbreeding. Our data present the possibility of another assumption that the two suspicious origin species of domestic ducks — mallard and spot-billed, may diverge after their divergence with domestic lineages or undergo rapid radiation, thus putting the origin time of the domestic lineage beyond the human civilization. We cautioned to explore this looks unlikely possibility and found that this phylogenetic relationship (Fig. 1e) naturally matched with data. All evidences including PCA, IBS tree, genetic structure, f_3 -statistic and species-specific alleles' distribution combined to implied the new

evolutionary relationship between mallard, spot-billed and domestic lineages (Fig. 1e right) is the more compatible topology than traditional one. Further *ada*i simulation and PSMC analyses, which rely on different rationale and different aspects of data, derived coincident divergence times and indicated similar demographic history, supporting our inference. In fact, some research on domestication has found similar result with ours on divergence time between domestic lineage and its closest wild relative. But they may either narrow the inferred divergence time to plausible range or set stringent limitation on ‘domesticated time’. An example we found is [6] using same methodology (*ada*i simulations) in dog. Their simulation actually came out with two divergence time of best fit between wolf and dogs, whereas they discarded the longer one (160~250kya) which is similar to our results simply due to it looks unlikely in biology. Our case with two closest wild relatives provided an opportunity to avoid directly inferring divergence time that may be confounded by mutation bias and other factors. Thus, instead of inferring exactly divergence time, we turned it into determining on two alternative phylogenetic topologies, since both models surfing from the same mutation rate or other bias thus wouldn’t influence the comparisons. Consequently, the scale of divergence time between domestic and wild was largely established on this topology and self-contained logic.

Although incomplete lineage sorting (ILS) and gene flow may confound the analyses, it’s prone not to cause a genome-wide consequence (e.g. The genomic average genetic distance between the mallard and spot-billed ducks (mean F_{ST} : 0.0637) significantly lower than that in mallard/spot-billed vs. domestic pairs (mean F_{ST} : 0.2832, 0.2498) (Fig. 4b)). we emphasized that it is remain unclear how domestication process was evolved and which is the major force, nature or human’s intentional selection. As the domestic may have hybridized with wild counterparts, thus during the long history

from their divergence, genomes of both lineages may continually suffer from introgressions. Some further works may need to take, such as utilizing emerging chromosome level duck pangenomes to detailly distinguish the evolutionary history of each genome segment, revealing the phylogenetic discordance for each genomic region, distinguish the ILS/hybridization to provide high resolution of tree topology across the genome and involving more widespread breeds/species samples. The reason for the fragmented single of ancient selection analyses in wild lineages may due to the misidentification of ancestral alleles, as both outgroup – Muscovy duck and Taihu goose, are domestic that may suffer the same genetic drift as domestic ducks. So further work using more appropriate outgroup such as bar-headed goose may improve ancient selection analysis, as well as $\partial a \partial i$, though ancestral allele is not must be deduced in $\partial a \partial i$.

Meanwhile, functions of genes under selection could provide information to understand domestication process, we also propose that it could be more efficient, especially to unrevealed fundamental shift associated with the domestication, in the context of a clear profound knowledge of demography[7]. Recent research have suggested that the critical changes at only a few domestication loci may not true, instead of many mutations of small effect, especially those frequency changes at regulatory regions play much more prominent role [2, 4, 7, 9]. The long-term domestication process further supported this view and highlight the evolution context for those functional studies. Similar situation can be found in the flightless research of different mechanisms underlying different cases [18, 50]. Our research unexpected found the divergence time between domestic duck lineages and their closely wild relatives surprised longer than the origin of human civilization may provide an excellent case to understand domestication process. We proposed that the important

phenotypic difference between wild and domestic duck, long distance flight abilities, may also a long-term evolutionary process that start from their divergence. It further implied the importance of evolutionary context on functional study and the complexity of flight evolution. Flight certainly seems to be linked to evolutionary success and a hard process [51, 52]. Birds' flight originated from dinosaurian long time ago and made it through the extinction, and evolved to different extent of flight ability. 'Most extinct birds are flightless' might be the interpretation for the decline of domestic lineage and the maintain of wild population, especially when it remains unclear whether the domestic ducks lost flight or not successfully acquire[52-54] (Fig. 2d). It may suggest the domestication of duck was more a natural consequence related to environment and resource rather than singly human dominance [8].

In addition, we found that duck domestication was accompanied by selection at genes affecting neuronal activities (e.g., *SH3GL2* [6](Wang et al., 2013)) and lekking behaviour (e.g., *HEXB*, *B4GALT1* and *HSD17B4*). Moreover, genes associated with lipid, carbohydrate, and protein metabolic processes and food intake control were under selection (e.g., *RMI1*, *MRAP2* and *MC4R* [4](Rubin et al., 2012)). In summary, an adaption to food-rich and artificial feeding environments with the development of agriculture may have contributed to duck domestication.

The rapid speciation in mallard and spot-billed, that we have demonstrated the speciation may occur within only 70 Kya, provides an excellent resource for speciation studies. Some genes located in differentiation region may contain importation clue to the driving force of the speciation process that we will further investigate. All "divergent peaks" located in scaffolds homologous to the 'Z' chicken chromosome are similar to the situation in flycatchers. Male plumage traits of the divergence species (mallard/spot-billed duck or collared/pied flycatcher) are quite

different, and for flycatcher, both species-specific male plumage traits and species recognition are located on the Z chromosome [55, 56]). Linked selection is suggested to dominantly drive the evolution of genomic differentiation rather than gene flow [3, 34, 37]. The lack of elevated d_{xy} in “differentiation islands” and other analyses supported this suggestion. Furthermore, Tajima’s D is not reduced in differentiation islands potentially due to new strong positive selection (Fig. 4; Fig. S21). Tajima’s D is used to detect the selection of new mutations and correlated to the difference between π and the expectation of π , which is related to the number of SNPs [57]. Due to the short divergence time, the mallard and spot-billed did not deposit many new mutations that increased population mutation rate (θ_w) after selection (new mutations increase diversity at a low frequency thus would mainly increase the θ_w rather than nucleotide diversity (π) to reduce the Tajima’ D). Thus, the results in the allele frequency spectra did not skew towards rare alleles, whereas the collared and pied flycatchers (approximately 340 Kya-2 Mya) and Darwin’s finches show reduced Tajima’s D in their genomic divergent islands due to the long-time interval after initial selection, which may be associated with the driving force of speciation [31, 37]. In the end, we note that evidence in the genetic structure showed that mallard and spot-billed (especially the spot-billed duck) have interspecific similarities to domestic ducks (Fig. 1d and Fig. S25; especially when $K=3,4$). This finding may pose a significant threat to the genetic diversity of wild stocks.

Conclusions

By taking advantage of massive whole genome sequencing and experimentally validating the activity of selected mutations, our research put the origin of domestic duck lineages back to a time far beyond the human civilization, and delved into their potential functional difference from wild counterparts in genomes. The shorter

divergence time between mallard and spot-billed ducks serves as supporting evidence for such an unexpected long timeframe of domestic lineages, is further indirectly substantiated by the special features and landscape of very small genome islands. Thus, our exploration opens up new avenues for understanding the animal domestication that would benefit future studies of this field when taking this perspective shift into account. The selection and speciation results represent an initial opportunity to discover the accompanying functional evolution, especially in wild ducks, and the formation of genome islands and its relation to speciation.

Methods

Sample collection

A total of 15 mallard, 14 spot-billed and 8 other species wild ducks (with hunting permission granted by the local government,) together with 8 domestic duck breeds of 114 individuals (Shaoxing duck from [58]) and two outgroup species (10 individuals pool each) were analysed in this study. The geographical distribution of samples collection and sequencing detail could be found in Fig. S1 and Table S1. All wild ducks were live-trapped far from human activity area to ensure its wildlife. Including 9 mallard (9/15) and 8 spot-billed (8/14) from Fenghua, Zhejiang province (121°39'E 29°67'N); 6 mallard ducks (6/15) and 6 spot-billed (6/14) from Hangzhou, Zhejiang province (120°21'E' 30°25'N). 8 other species wild ducks were collected from Poyang late, Jiangxi province (116°59'E 28°96'N) and Fenghua, Zhejiang province, separately. Fresh blood was obtained from wing veins and rapidly frozen under -20 °C. All experiments with birds were performed under the guidance of ethical regulation from Animal Care and Use Committee of China Agricultural University,

Beijing, China.

Assembly and annotation

The assemble of the two wild ducks were performed using the similar pipeline as Beijing duck assembly to exploit the specific genomic feature of wild or domestic duck and access the influence of genome assemble on subsequent variation analysis. Sequences from paired-end reads of 17 libraries for each individual with insert sizes ranging from 250 bp to 20,000 bp were assembled with SOAPdenovo and SSPACE software (Table S2). The super-scaffolds were positioned along the chromosomes based on the duck cytogenetic map and the comparative genomic map between duck and chicken (Yinhua Huang et al., 2013; Y. Huang et al., 2006). Reference gene sets for the mallard genome and the spot-billed genome were retrieved by merging the homologous set, the transcript set (predicted with 21 duck transcriptomes) and the *de novo* set (Yinhua Huang et al., 2013).

Whole-genome resequencing

Following autoclaving and standard phenol-chloroform extraction, we assessed DNA quality through agarose gel electrophoresis, measuring concentration and purity with Aglient 2100 and NanoDrop, selecting samples with >30 ng/uL and both 260/280 and 260/230 ratio within an ideal range that indicates minimal protein contamination. 30ug were required for each sample of wild species (13 spot-billed and 14 mallard, one gadwall, Baikal teal, pintail, falcated teal, common teal, wigeons each) and Pekin duck breed to construct paired-end libraries (400 bp) and sequence individually with 150bp reads. Other 6 breeds and 2 outgroups (muscovy ducks and Taihu geese) were pooled in equimolar quantities of DNA from 10 individuals each and sequenced two same lane each breed as technical comparison with 100bp paired-reads. Sequencing was conducted according to the instructions of Illumina HiSeq platform carried out by

BGI(Shenzhen). The different sequence strategy makes us take additional steps to process data to eliminate its influence. Which makes our article a little bit miscellaneous and difficult to follow.

Alignment, filters and SNP calling.

Clean reads were aligned to Pekin duck assembly (BGI_duck_1.0) using the bowtie2 with default parameters, resulting in the following metrics:

average depths across the genome :50X, 49X, 15X, 22X

alignment rates: 0.90, 0.64/0.82, 0.87/0.88/0.90, 0.88

genome coverages: 0.96, 0.92/0.94, 0.96, 0.95. (percent of the genome is covered at least by one read.)

Above metrics were obtained for average of domestic breed pools, Taihu goose/Muscovy pool, average of mallard/spot-billed/Pekin individuals respectively or totally, and average of other wild individuals. Duplicated reads were removed individually from samples using Picard v. 1.119 and realignment process are performed before SNP calling. After removing reads that map to multiple places, we called SNPs using GATK v. 3.11 and filtered SNPs according to the coverage with thresholds of $\leq$ one third of the mean or $\geq$ three-fold of the mean and MQ value $\leq$ 28 [59]. SNP used in IBS, PCA and genetic structures were called from multiple bam samples by using joint variant calling mode of GATK. We merged individual bam files into one population bam sample and jointly call SNPs with pool samples, or random sample bam reads into same coverage when necessary.

Linkage Disequilibrium (LD), Identical By State (IBS), Principal Component Analysis (PCA), f3-statistics and population structure

LD level in Pekin, Shaoxing, mallard duck and spot-billed duck populations were measured through calculating the correlation coefficient (r^2) of alleles using the

PLINK v. 1.07[60].The parameters were set as follows: --ld-window-kb 500 --ld-window 99999 --ld-window-r2 0 --r2. To reduce effect of LD, SNPs were pruning by LD and subsequently used for the PCA, f3-statistics and population structure analyses (Fig. S12). We constructed phylogenetic trees by using neighbor-joining method with MEGA based on IBS distance matrix data of all individuals calculated by the PLINK using the above sampling SNPs[61]. PCA was performed using the GCTA64 v. 1.24[62]. The f3-statistics were calculated using the ADMIXTOOLS v5 and genetic structure was inferred using the admixture program.

$\partial a \partial i$ simulation and PSMC analysis

$\partial a \partial i$ v. 1.6.3 utilize the joint allele frequency of populations to simulate its changes through history under given demographical model. SNPs frequency of mallard, spot-billed and other 8 domestic populations for $\partial a \partial i$ were obtained from each of the 10 VCF files, then combined by using a way like outer join of SQL. (i.e. include SNPs that were monomorphic within one population but variable in their combined samples (10 populations); <https://github.com/ruiliuacd/duckdomestic1>). After LD pruning, We then inferred ancestral alleles by using Muscovy ducks and Taihu geese as outgroups using the following thresholds: (1) more than 30-fold coverage when SNP fixed in Muscovy or both outgroups; and (2) the both outgroups were fixed the same directions. Subsequently, we quantized the frequency of each SNP of 8 domestic duck populations into one population and simulated the divergence among spot-billed, mallard and “domestic” ducks using split without migration, split with migration and IM models with 50 bootstraps each (Additional file 1 section 5.5). N_{ref} was calculated

with the following formula:
$$N_{ref} = \frac{\text{Watterson's estimator } \theta}{4 * \text{mutation rate} * L}$$
; and L represents the length of the SNPs from and diluted by the rate of SNP sampling.

$\partial_a \partial_i$ use a diffusion approximation to Wright-Fisher process to simulate forward time whereas PSMC use a different theoretical base, coalescent, that utilize the change of density of mutations along individual genome. In both PSMC ([10, 63] Li & Durbin, 2011; Lohmueller, Bustamante, & Clark, 2010) and $\partial_a \partial_i$ simulations, We assumed that the mutation rate per year was 9.97×10^{-10} and the generation time was one year [64].

Positive selection in early ancestors of mallard and spot-billed ducks

To detect selection that occurred in wild duck lineage (mallard and spot-billed) during their ancestor in conjunction with, or shortly after their divergence from the domestic lineage, we used a method derived from [39] that is particularly suited to detect older selective sweeps during or shortly after divergence. We selected SNPs that derived allele existed (including fixed) in mallard or spot-billed. Ancient sweeps were enriched mainly through searching for genomic regions with dearth of derived allele shared with domestics among wilds. (i.e. selected and linked SNPs reached high frequency to fixed. Whereas the allele frequency spectrum of those regions will be recovered by wild duck alleles, that is not shared with domestic lineages, with frequency spectrum towards normal to higher frequency since age old). This method will naturally disregard regions of recent selection or purifying selection in wild because those regions with derived allele skewed towards lower frequency wild alleles and making lower predicted domestic frequency thus leading weak signal (Additional file 1).

We calculated the derived allele frequency of 18,083,967 selected SNPs in both the wild and domestic lineage. As expected, they were highly correlated (Fig. S22a) This correlation was used to calculate the expected derived allele frequency in the domestic lineage as a function of the frequency in the wild lineage. log ratio of the sum of

observed/expected derived frequency in 20-kb windows with a step size of 10 kb were applied to scanning. After Z-transforming the genome-wide distributions, we arrived at measure S and defined selective sweep regions of the early mallard and spot-billed ancestors using the thresholds of -0.2819 (4-fold standard deviations away from the mean) for autosomes and -0.3089 for sex chromosomes (1.5-fold standard deviations away from the mean). Recent selective sweep regions in the domestic lineage could also reduce the S value; thus, we excluded regions that show lower zHp (CDRs) in the domestic lineage.

Selecting candidate casual mutation to verify its functional activity

For the sake of simplicity, we search for the regulation regions that located on the candidate selected regions. We categorized the selection regions based on their overlap with CDS, UTR, intron, intergenetic regions. And rank SNPs by the ΔAF between wild lineage and domestic lineage. we selected the SNPs that located in UTR, intron, intergenetic region and closed to the genes that may play important functional in the duck evolution or domestication. This experiment can also be viewed as SNP validation pruning see Additional file 1.

Abbreviations

PCA: Principal component analysis

LGP/LGM: Last glacial period/maximum

PSMC: Pairwise sequential Markovian coalescent

Kya/Mya: Kilo/Million years ago

LD: Linkage disequilibrium

ILS: Incomplete lineage sorting

CDR: Candidate domestication region

IBS: Identical by state

F_{ST}: Fixation index/population differentiation

H_p: Pooled heterozygosity

ΔDAF: Difference of Derived Allele Frequency

dxy: Average number of pairwise differences between sequences from two populations

Declarations

Ethics approval and consent to participate

All experiments with birds were performed under the guidance of ethical regulation from Animal Care and Use Committee of China Agricultural University, Beijing, China.

Consent to publish

All of the authors agreed for publication

Availability of supporting data and materials

The duck genome resequencing data are deposited under the BioProject PRJNA315043, Taihu goose and Muscovy duck data were under PRJNA313495. The genome assembly of a mallard (ASM222489v1) and a spot-billed has been deposited in GenBank PRJNA392350.

Competing interests

The authors declare no completing interests

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(2017SKLAB06-2).

Author contributions

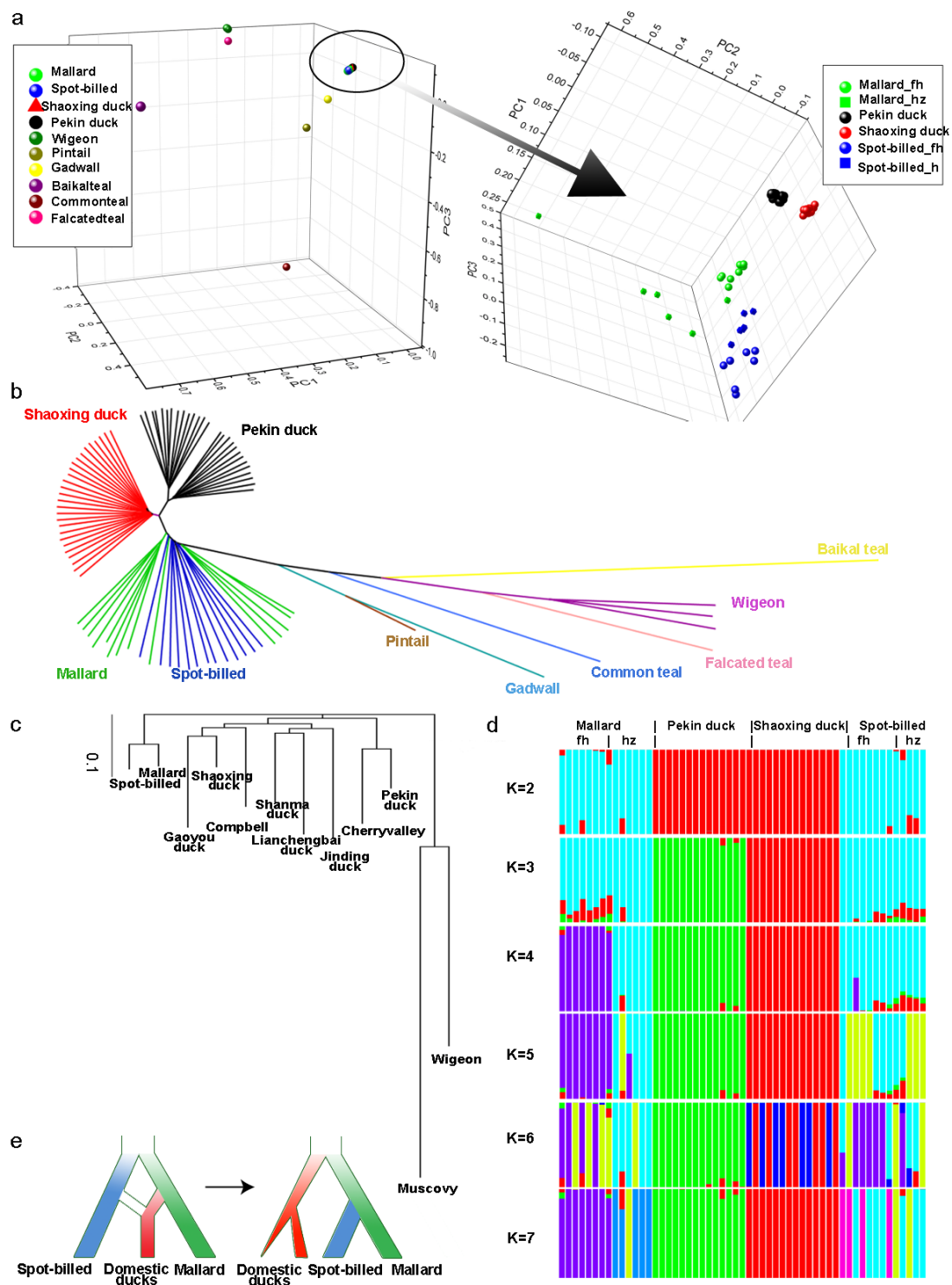
N.L. and Y.H.H. designed the project. Z.L.L., H.F.L., L.C., H.B.C., W.T.S., E.G.R and R.L collected and purified DNA samples. R.L performed population genetics analysis. E.G.R. and R.L. performed luciferase activity analysis. R.L. and Y.H.H. wrote the manuscript. Q.W.L., Y.Z, G.Y.F., H.F.L., Y.S.S., W.B.C., X.L. and X.X. assisted in analysis, pictures and material's supporting and manuscript's writing. R.L. and Y.H.H. revised the manuscript.

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692

693 **Figure legends**



694

695 **Figure 1: Inference of phylogenetic topology in wild and domestic ducks. Pekin**

696 and Shaoxing ducks representing domestic ducks were used in the principal

component analysis (PCA), phylogenetic analysis based on identity-by-state (IBS)
 and genetic structure analyses. “_fh” and “_hz” mean spot-billed and mallard
 collected from Fenghua and Hangzhou, respectively. **(a)** PCA plot constructed for
 PC1 (Principal Component 1), PC2 and PC3 using 10,525 high-quality filtered SNPs
 from 27 Pekin and 27 Shaoxing ducks and 35 wild ducks (one Baikal teal, one
 gadwall, one pintail, one falcated teal, one common teal, three wigeon, 13 spot-billeds
 and 14 mallards) shows that domestic ducks are close to spot-billed and mallard wild
 ducks. **(b)** Neighbour-joining tree based on IBS distance of 80,505,576 SNPs. Pekin
 duck samples are from the lines (Pekin duck are from high/low growth lines See
 Additional file 1). **(c)** Neighbour-joining tree based on genome wide pairwise fixation
 index (F_{ST}) of 44,852,612 SNPs. **(d)** Genetic structure analyses of 13 spot-billed and
 14 mallard wild ducks along with 14 Pekin and 14 Shaoxing domestic ducks using an
 admixture ranging K from two to seven. **(e)** Inferred demographic models for wild and
 domestic ducks inferred based on the genetic analyses of **a-d** (right). The traditional
 demographic model, which hypothesizes that domestic ducks originated from
 mallards or the descendant of a mixture of spot-billed and mallard ducks (left), is
 incompatible with observations from the neighbour-joining trees **(b-c)** and genetic
 data **(d)**.

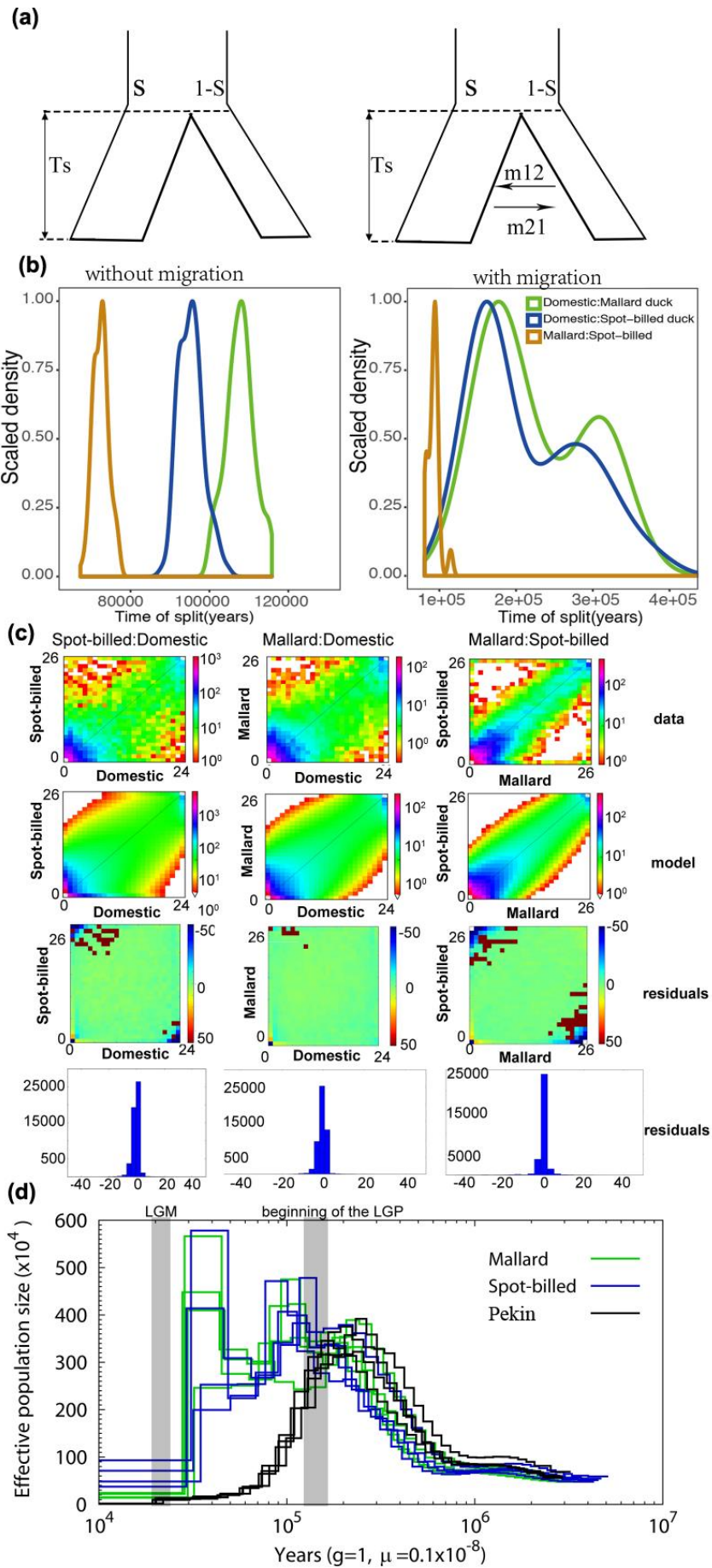
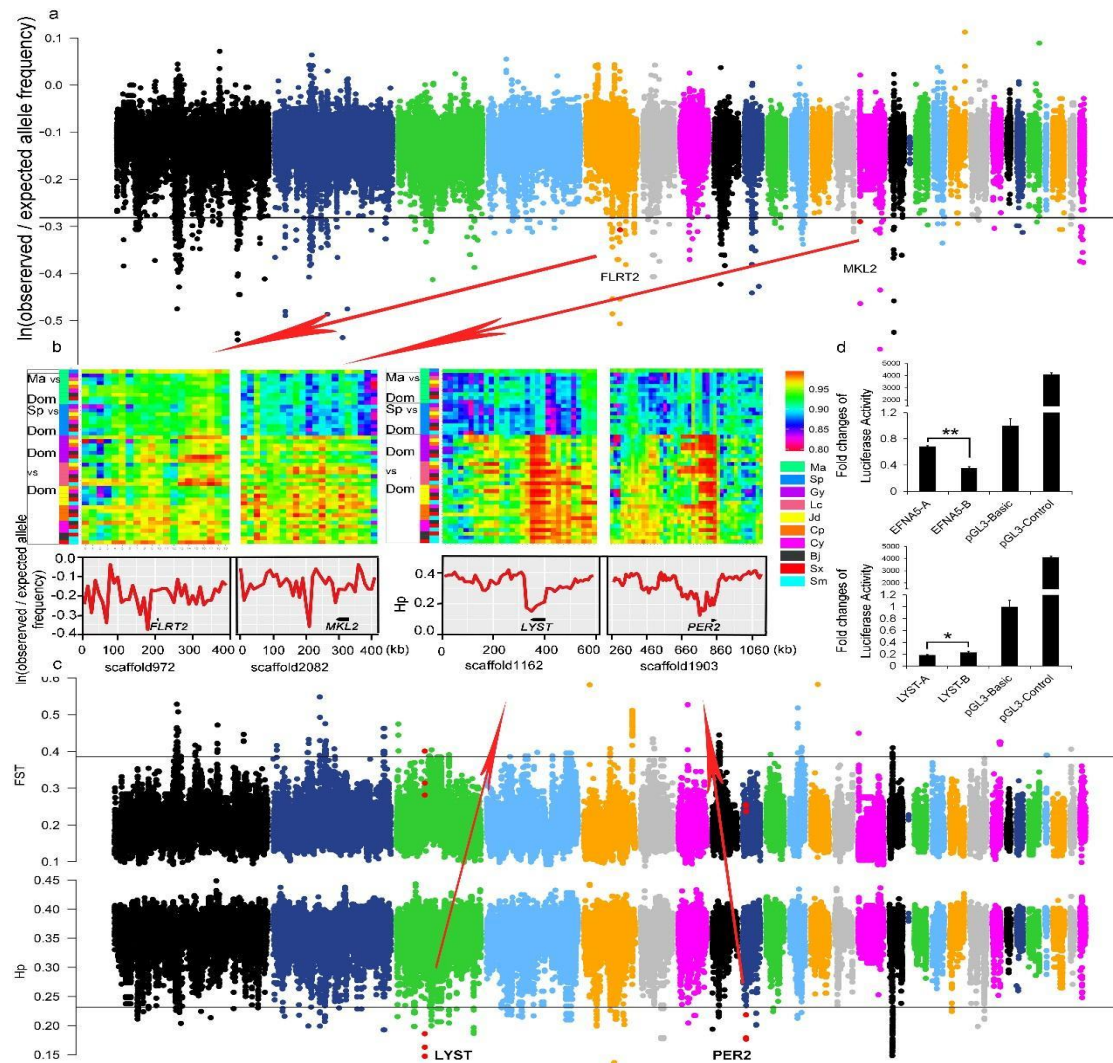


Figure 2: Inference of joint demographic history of ducks. Wild ducks included spot-billed and mallard ducks. Domestic ducks included Pekin, Cherry Valley, Campbell, Gaoyou, Jingding, Lianchengbai, Shaoxing and Shanma. **(a)** Illustration of the two models (with/without migration) used in the history inference through the $\partial a \partial i$ analysis. Parameters corresponding to the code are provided in the supporting information. **(b)** Distribution of estimated T_s (divergence time) of each model in 100 simulations between spot-billed and mallard (orange), mallard and domestic lineage (green), spot-billed and domestic lineage (blue) corresponding to model with/without migration. **(c)** Joint Site Frequency Spectrum (SFS) for observed data, simulation data and residuals. **(d)** Demographic history inference of mallard duck, spot-billed duck and Pekin duck through the PSMC analysis.



729
730 **Figure 3: Summary of selective sweep analyses of the early wild duck lineage and**
731 **domestic duck.** (a) Distribution of $\ln(\text{observed}/\text{expected derived allele frequency})$
732 calculated in a 20 kb window with a sliding size of 10 kb along scaffolds that are
733 arranged according to their alignment to chicken autosomes. The horizontal black line
734 indicates the threshold at $\ln(\text{observed}/\text{expected derived allele frequency})=-0.3136$
735 (mean of $\ln$ values - 4 standard deviations). Outlier regions that were visualized in

736 detail by identity scores (IS) and re-computing values in smaller window sizes in **(b)**
 737 are highlighted with red dots. **(b)** Details of four putative selective sweep regions
 738 around the *FLRT2*, *MKL2*, *LYST* and *PER2* genes. The top of each panel shows the
 739 degree of haplotype sharing in pairwise comparisons among populations. Coloured
 740 boxes (left) indicate the comparison performed on that row. “Ma,” “Sp,” and “Dom,”
 741 represent mallard, spot-billed and domestic duck, respectively. The y-axis (downside)
 742 of *FLRT2* and *MKL2*, which are associated with retina/heart morphogenesis under
 743 selection in wild ancestors, are $\ln(\text{observed/expected derived allele frequency})$,
 744 whereas the y-axis of *LYST* and *PER2*, which were under selection in domestic duck,
 745 are H_p . **(c)** Distribution of F_{ST} between the wild and domestic duck and H_p in the
 746 domestic duck, calculated in a 40 kb window with a sliding size of 20 kb along
 747 scaffolds that are arranged according their alignment to chicken autosomes. The
 748 horizontal black line indicates the threshold at $F_{ST}=0.3858$ or $H_p=0.2319$ (mean of $\ln$
 749 values - 4 standard deviations). Red dots indicate the location of outlier genes (4
 750 standard deviations away from h_p means) that were visualized in detail in **(b)**. **(d)**
 751 Comparison in luciferase activity between wild-type and domestic-type in DF1
 752 (chicken embryonic fibroblast) for alleles of *EFNA5* and *LYST* that is putatively
 753 selected in wild (spot-billed and mallard) and domestic (Pekin) duck cells. “**A**” and
 754 “**B**” represent wild duck and domestic duck, respectively. “*” $P \leq 0.05$.

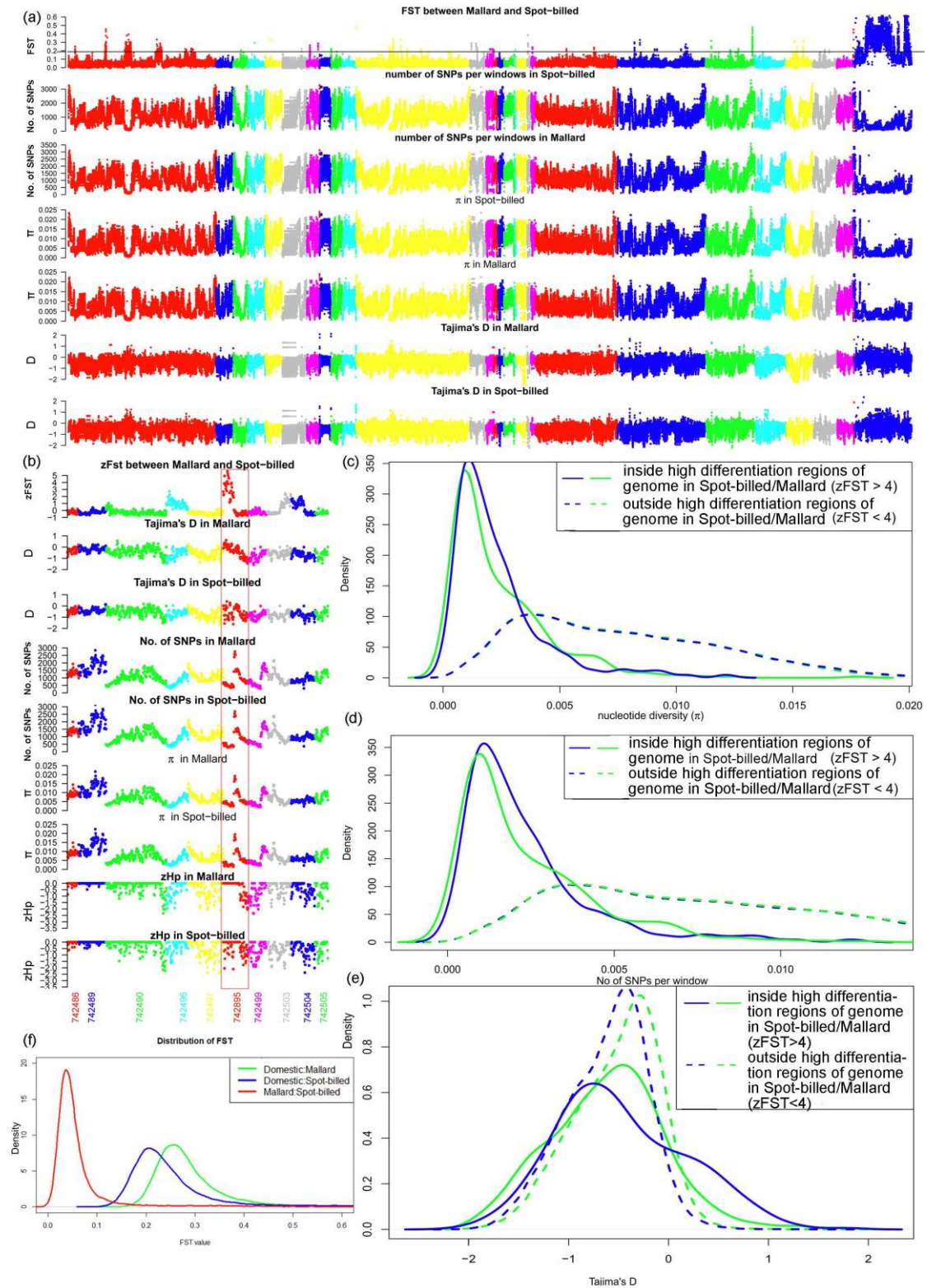


Figure 4: Genomic landscape of species divergence between spot-billed and mallard ducks. (a) Distribution of population genomic parameters of the Pekin duck assembly, in 40 kb windows with sliding 20 kb steps. The blue scaffolds in the right were referred to as duck Z scaffolds. The black horizontal line indicates the threshold

at $F_{ST}=0.1888$ (mean of F_{ST} + 4 standard deviations, referred to as $Z(F_{ST})=4$). The plots show the F_{ST} between spot-billed and mallard, number of SNPs per window, nucleotide diversity (π), and Tajima's D for spot-billed and mallard. **(b)** Distribution of population genomic parameters along ten example scaffolds; this distribution is part of (a). **(c-e)** Distribution of nucleotide diversity (π), number of SNPs per window and Tajima's D inside/outside high differentiation region of autosomes (high differentiation regions, i.e., $zF_{ST} \geq 4$, corresponding to bins above the horizontal line in (a) autosomes) in mallard and spot-billed ducks. **(f)** Distribution of pairwise F_{ST} for 40 kb windows with sliding 20 kb steps. The red line shows the plot in the first row of **(a)**. Blue and green lines show the pairwise F_{ST} between spot-billed/mallard and domestic ducks (combining 8 domestic breeds).

772

773 Tables

774 **Table 1. Tests for population mixture of mallard, spot-billed, Pekin and**
 775 **Shaoxing ducks**

Source 1	Source 2	Target	f3	Standard error	Z	SNPs
mallard	spot-billed	Pekin	0.007886	0.000748	10.546	9,997
mallard	spot-billed	Shaoxing	0.008662	0.000814	10.643	9,943
mallard	Spot-billed	Mix Pekin, Shaoxing	0.007279	0.000305	23.834	2,930,914
mallard_hz	spot-billed_hz	Pekin	0.008656	0.001076	8.045	9,975
mallard_hz	spot-billed_hz	Shaoxing	0.010200	0.000683	14.930	9,907
mallard_fh	spot-billed_fh	Pekin	0.007320	0.000603	12.147	9,986
mallard_fh	spot-billed_fh	Shaoxing	0.007380	0.000988	7.470	9,921

776 Considering the extensive substructure exist in wild ducks which reflect those individuals may trace
 777 their recent ancestry to different populations even species, we assume different clusters combinations
 778 to source 1 and source 2, and combine different domestic breeds to one target population, to minimize
 779 the population effect. All the Z were significantly greater than zero.

780

781

Table 2. Top 32 enriched biological process terms among genes under positive selection in the early spot-billed and mallard ancestors

Gene ontology term	P-value	Number	Size
positive regulation of tyrosine phosphorylation of Stat3 protein	6.29E-05	4	12
regulation of activin receptor signalling pathway	0.000382	2	2
positive regulation of cytosolic calcium ion concentration involved in phospholipase C-activating G-protein coupled signalling pathway	0.000387	3	8
detection of virus	0.001131	2	3
D-aspartate import	0.001131	2	3
metanephros morphogenesis	0.002232	2	4
L-glutamate import	0.002232	2	4
negative regulation of extrinsic apoptotic signalling pathway in absence of ligand	0.003441	3	16
positive regulation of granulocyte macrophage colony-stimulating factor production	0.003672	2	5
respiratory system development	0.003672	2	5
determination of left/right symmetry	0.003701	5	53
spinal cord motor neuron differentiation	0.004119	3	17
lung morphogenesis	0.004119	3	17
ionotropic glutamate receptor signalling pathway	0.004872	3	18
ventricular cardiac muscle tissue morphogenesis	0.004872	3	18
dorsal/ventral pattern formation	0.005176	4	36
negative regulation of peptidyl-threonine phosphorylation	0.005437	2	6
serotonin metabolic process	0.005437	2	6
lung epithelium development	0.005437	2	6
cytokine-mediated signalling pathway	0.006291	4	38
nucleotide-excision repair	0.006613	3	20
neutrophil chemotaxis	0.006613	3	20
negative regulation of epithelial cell proliferation	0.006904	4	39
positive regulation of heart rate	0.007514	2	7
inositol phosphate-mediated signalling	0.007514	2	7
bud elongation involved in lung branching	0.007514	2	7
limb morphogenesis	0.009839	3	23
long-chain fatty acid metabolic process	0.00989	2	8
dicarboxylic acid transport	0.00989	2	8
interleukin-6-mediated signalling pathway	0.00989	2	8
thrombin receptor signalling pathway	0.00989	2	8
response to wounding	0.012416	3	25

Full lists of enriched terms are presented in Table S15. Terms considered to be associated with the enhancement of long-range flight ability are shown in bold. Size represents the total number genes of this term.

Table 3. Mean values of genomic parameters inside/outside high F_{ST} (between mallard and spot-billed ducks) regions in autosomes

Species		pi (π)		No. of SNPs		Tajima's D	
		mean	sd	mean	sd	mean	sd
Mallard	Inside	0.002421	0.00217898	360	305	-0.61100	0.579419
	Outside	0.007595	0.00418171	1,134	569	-0.52064	0.433172
Spot-billed	Inside	0.002331	0.00196792	334	280	-0.47211	0.656325
	Outside	0.007683	0.00430549	1,161	597	-0.6228	0.415501

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